

# Analytical Report on `main.md` and `prompt.md`

## Executive summary

Both local files were available and sufficiently detailed for a full review. Taken together, they are strongly aligned: `prompt.md` asks for a deep research and paper-framing exercise around a drone-based individual tree phenology project, and `main.md` confirms that the repository's real centre of gravity is not stand-alone crown detection, but repeated UAV surveys, orthomosaic alignment, graph-based crown identity tracking, consensus crown generation, crown-level RGB phenology signals, field/QField linkage, and only then downstream satellite or species-classification experiments.

[filecite:turn0file1](#) [filecite:turn0file0](#)

The clearest high-confidence conclusion is that the work should be framed as a **tree-identity-preserving temporal phenotyping system from repeated UAV RGB surveys**, with crown delineation as an enabling stage and satellite/DINOv2 work as downstream or secondary. That framing is exactly what the prompt hints at, and it is explicitly supported by the main project map, which states that “the hard problem is then temporal identity” and that “the final scientific object is not just a crown detection. It is a temporally stable tree identity”. [filecite:turn0file1](#) [filecite:turn0file0](#)

The most robust paper contribution is therefore not “we ran Detecttree2 on orthomosaics”, because that alone would look incremental in a literature where deep-learning crown delineation is already well established. Instead, the strongest contribution is the integrated method that links repeated orthomosaics, multi-threshold crown stores, translation-focused post-alignment, graph-based temporal association, gap-filling, consensus crowns, and crown-level colour trajectories for phenology analysis. That framing is more defensible against the current individual-tree-crown literature, which is dominated by detection and delineation rather than tree-level longitudinal identity preservation; that latter point is an inference from the review literature and should be stated as such. <sup>1</sup>

The biggest weaknesses are also clear. The files do not surface a mature, fully specified longitudinal re-identification benchmark; early detection evaluation is reported, but tracking evaluation appears more qualitative or pipeline-oriented than paper-ready. There is also at least one threshold ambiguity in the phenology lane, and the files suggest that satellite and flowering work are more exploratory than core. Those issues do not invalidate the thesis. They simply mean the final framing should be **honest, narrower, and more methodologically disciplined**: centralise identity-preserving temporal crowns and crown-level RGB phenology, and treat satellite inference, species classifiers, and flowering as downstream extensions unless a stronger validation set is available. [filecite:turn0file0](#)

My overall recommendation is straightforward: pitch the thesis as **drone-based individual tree phenology monitoring**, pitch the paper as **graph-based temporal crown association plus consensus crowns for tree-level phenology from repeated UAV RGB orthomosaics**, and present satellite/DINOv2 experiments as a proof of downstream utility rather than the main novelty. That recommendation is consistent with the repository, with the prompt's own instincts, and with how the current literature would likely judge novelty and rigour. [filecite:turn0file1](#) [filecite:turn0file0](#)

## What the two files say and how well they match

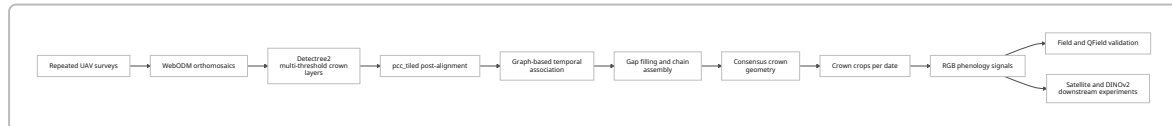
Two quotations capture the alignment unusually well.

“This is not just a detection project; the crown tracking / temporal identity / phenology-over-time aspect is the real core.” [filecite turn0file1](#)

“The final scientific object is not just a crown detection. It is a temporally stable tree identity...” [filecite turn0file0](#)

That is not a superficial overlap. [prompt.md](#) is effectively asking for a narrative that [main.md](#) already justifies technically. The prompt’s emphasis on repeated surveys, orthomosaic drift, crown boundary instability, graph-based tracking, medoid consensus crowns, phenology features, field/QField labels, and satellite follow-ons is all corroborated by the project map. [filecite turn0file1](#)  
[filecite turn0file0](#)

The repository workflow described in [main.md](#) also matches the literature-backed framing that would make this project publishable. Detectree2 is officially documented as a tree-locating system built on Detectron2’s Mask R-CNN architecture and designed for difficult dense-forest delineation tasks, while DeepForest remains a strong open-source canopy analysis library. At the same time, recent independent TLS-based validation work shows that hand-labelled aerial evaluation can substantially overstate performance for crown delineation, which strengthens the case for making **tracking, consensus geometry, and longitudinal stability** central rather than pretending crown segmentation alone is solved. [3](#)



This end-to-end chain is consistent with official photogrammetry and fieldwork tools. OpenDroneMap and WebODM explicitly foreground orthophotos, DEMs, GCP workflows, and relative-accuracy concerns; OpenDroneMap even separates relative/local from absolute/global accuracy and documents multi-temporal accuracy workflows. QField officially supports field data collection, GPS/GNSS use, digitising/editing, and project-based field workflows. Those official sources strengthen the credibility of the operational story already visible in the files. [4](#)

The main substantive mismatch is not between the two files, but between what the files can presently support and what a reviewer might demand. [main.md](#) documents the pipeline in impressive technical detail, yet it does not, at least in the material surfaced here, provide a complete published-grade quantitative benchmark for temporal identity tracking. That matters because the thesis is strongest when identity preservation is central. The implication is not that the framing is wrong; it is that the validation chapter must be sharpened around a labelled subset or carefully defined proxy metrics. [filecite turn0file0](#)

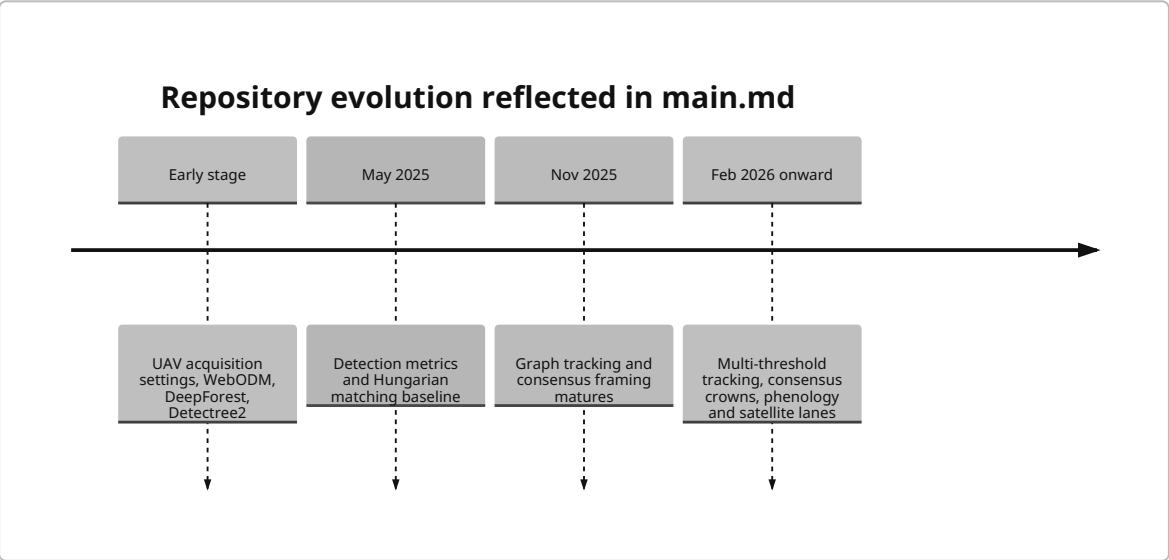
The grouped comparison below captures the core consistency position.

| Requirement area          | What <code>prompt.md</code> asks for                                                                                       | What <code>main.md</code> supports                                                                                                     | Assessment                                |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Core scientific problem   | Tree phenology at individual-crown level over time<br><a href="#">filecite:turn0file1</a>                                  | Temporal identity is explicitly called the hard problem; final object is a stable tree identity<br><a href="#">filecite:turn0file0</a> | Strongly consistent                       |
| Active technical pipeline | Orthomosaics, Detectree2, alignment, graph tracking, consensus crowns, RGB features<br><a href="#">filecite:turn0file1</a> | All of these steps are described as the current production pipeline<br><a href="#">filecite:turn0file0</a>                             | Strongly consistent                       |
| Field linkage             | QField / species / validation context<br><a href="#">filecite:turn0file1</a>                                               | Field/QField species annotations are included in the project map<br><a href="#">filecite:turn0file0</a>                                | Consistent                                |
| Satellite lane            | Important but possibly secondary<br><a href="#">filecite:turn0file1</a>                                                    | Satellite/GEE/DINO work is described as downstream and experimental<br><a href="#">filecite:turn0file0</a>                             | Consistent, should remain secondary       |
| Thesis or paper framing   | Decide whether this is enough for a thesis and how to pitch it<br><a href="#">filecite:turn0file1</a>                      | Historical reports show evolution from detection baseline to tracking and phenology pipeline<br><a href="#">filecite:turn0file0</a>    | Consistent, thesis yes; paper must narrow |
| Literature grounding      | Deep research and real citations where possible<br><a href="#">filecite:turn0file1</a>                                     | Main file names tools, baselines, reports, but not a built literature review<br><a href="#">filecite:turn0file0</a>                    | This report closes that gap               |

A second useful view is the internal project timeline reconstructed from `main.md`:

| Timepoint   | Evidence in <code>main.md</code>                                                                                                                                  | What it represents                            |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| Early stage | Early presentation with DJI Mini 4 Pro, WebODM, DeepForest, Detectree2, flight settings, and first tracking ideas<br><a href="#">filecite:turn0file0</a>          | Data acquisition and first detection baseline |
| May 2025    | Formal report with SIT polygon evaluation and Hungarian baseline matching; poor recall but decent IoU when correct<br><a href="#">filecite:turn0file0</a>         | Detection-only framing still dominant         |
| Nov 2025    | End-term material introduces graph ideas, medoid/ intersection/union-shrink discussion, and a more mature tracking framing<br><a href="#">filecite:turn0file0</a> | Tracking becomes the conceptual centre        |

| Timepoint       | Evidence in <code>main.md</code>                                                                                                                                                             | What it represents                               |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| Feb 2026 onward | Mid-term and March notebooks move to multi-threshold stores, alignment, chain assembly, consensus crowns, phenology features, and satellite experiments<br><code>filecite[turn0file0]</code> | Current end-to-end temporal phenotyping pipeline |



Sequential response to the requirements in `prompt.md`

What follows addresses each item in `prompt.md` in order, using the repository map in `main.md` as the consistency check.

A. Suggested paper or thesis title options

“Suggested Paper/Thesis Title Options” `filecite[turn0file1]`

“The hard problem is then temporal identity...” `filecite[turn0file0]`

This requirement is fully supportable, and the file cross-check is decisive: titles should foreground **repeated UAV RGB surveys, temporal crown identity, graph association, consensus crowns, and phenology** rather than generic crown detection. Titles that emphasise only Detectree2 or crown delineation would understate the real contribution and invite the “just applied an existing tool” criticism. That conclusion is reinforced by the literature, where crown delineation itself is already a crowded area.

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B. Abstract drafts

“Abstract Drafts” `filecite[turn0file1]`

“The final scientific object is not just a crown detection. It is a temporally stable tree identity...” `filecite[turn0file0]`

Also fully supportable. The correct abstract should say that repeated UAV RGB orthomosaics are easy to produce, but stable crown identity through time is hard because of drift, distortion, and segmentation inconsistency. It should then position the contribution as a pipeline that combines multi-threshold crown delineation, post-alignment, graph-based association, consensus crowns, and crown-level RGB phenology signals. It should **not** claim canopy-height modelling as a present central result, because `main.md` explicitly notes that the active pipeline does not yet compute canopy height even though the companion paper context mentions DTM plus DSM. [filecite:turn0file0](#)

### C. How to frame the thesis or paper

“What should the central thesis / paper framing actually be?” [filecite:turn0file1](#)

“The final scientific object is not just a crown detection...” [filecite:turn0file0](#)

This is the single most important item, and the answer is clear. The central framing should be: **an end-to-end method for constructing temporally stable, crown-level phenology records from repeated UAV RGB orthomosaics**. That statement is both faithful to the repository and more novel than a plain crown-detection paper. It also fits the current literature position: instance delineation from aerial imagery is mature and competitive, but longitudinal, identity-preserving individual-tree phenology from repeat orthomosaics is much less standard; that second clause is an inference from the review literature and from the kinds of papers that dominate the field. [6](#)

### D. Related work map

“Related Work Map” [filecite:turn0file1](#)

“The full conceptual chain is... crown delineation... graph-based temporal crown tracking... phenology features... satellite/GEE/DINO feature extraction” [filecite:turn0file0](#)

This requirement is fully supportable, and the correct map has five buckets. First, crown detection and delineation: Detectree2, Detectron2/Mask R-CNN, DeepForest, and broader ITCD literature. Second, registration and orthomosaic alignment: phase cross-correlation and higher-level registration literature. Third, temporal association and object identity: graph-based data association and plant-level temporal reuse pipelines. Fourth, phenology signal extraction from imagery: RGB-based greenness and image-derived phenological status. Fifth, downstream satellite and foundation-model features: Sentinel-2, Earth Engine, DINOv2. [7](#)

### E. Contribution and novelty assessment

“Is there enough here for a thesis? What is actually novel?” [filecite:turn0file1](#)

“The hard problem is then temporal identity...” [filecite:turn0file0](#)

Yes, there is enough for a thesis if the novelty is pitched correctly. The strongest novel contribution is **not** inventing a new segmentation backbone. It is solving an ecologically meaningful, practically messy longitudinal problem by integrating repeated UAV capture, crown delineation, residual alignment, graph-based association, gap recovery, consensus geometry, and per-crown temporal phenology descriptors in one coherent pipeline. The literature supports why that is a real contribution: delineation is hard, evaluation is uncertain, and even strong aerial models can underperform under stricter or independent validation. [8](#)

## F. Responses to methodological doubts

"How should I justify the use of orthomosaics / Detectree2 / graph matching / medoid crowns / rule-based phenology scores / downstream satellite classifiers?"

filecite turn0file1

"Use `pcc_tiled` for alignment by default." and "current code uses medoid by default."

filecite turn0file0

These methods are defensible, but not all of them should be equally central. Orthomosaics are practical because they make spatial crown association and GIS-linked field validation tractable; official ODM/ WebODM documentation supports them as standard outputs and explicitly discusses accuracy management. Detectree2 is defensible because it is an instance-segmentation method specialised for tree crowns and built on a proven Mask R-CNN lineage. Phase cross-correlation is defensible because it provides efficient subpixel translation registration and is a sensible post-hoc correction when residual translational shifts dominate. Medoid consensus crowns are defensible because they produce a stable geometry from noisy per-date polygons. By contrast, rule-based deciduousness scoring is better framed as a **heuristic phenology signal or baseline**, not as definitive biological truth unless a stronger label set exists. Satellite classifiers are best framed as **downstream utility demonstrations**. 9

## G. Anticipated reviewer criticism

"What will reviewers likely criticise?" filecite turn0file1

May 2025 detection metrics show low recall despite decent mean overlap when correct.

filecite turn0file0

This is answerable with high confidence. Reviewers are most likely to attack five things: weak or incomplete tracking ground truth; overclaiming segmentation quality; reliance on orthomosaics despite stitching artefacts; under-validated rule-based phenology labels; and an over-broad mixture of drone, field, and satellite stories. Literature strongly supports these concerns. Recent UAV work shows orthomosaics are useful but can be compromised by stitching artefacts, and recent TLS-based evaluation shows crown delineation performance can look much worse when validated independently.

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## H. Strong paper structure

"What is the strongest paper structure?" filecite turn0file1

`main.md` already organises the project as UAV acquisition, crown detection, alignment, graph tracking, consensus crowns, phenology, field validation, downstream satellite experiments. filecite turn0file0

Fully supportable. The strongest structure is pipeline-first and contribution-driven: problem and motivation; related work; study design and repeated UAV data; crown delineation; orthomosaic post-alignment; temporal graph construction; consensus crowns; phenology features and labels; validation and ablations; discussion and limitations; downstream utility. Satellite experiments should be moved near the end or into a dedicated "downstream utility" section. That preserves conceptual coherence.

filecite turn0file0

## I. Thesis framing versus paper framing

“Should the thesis be broader than the paper?” [filecite\turn0file1](#)

`main.md` includes archived notebooks, historical exploration, and downstream packages beyond the active pipeline. [filecite\turn0file0](#)

Yes. This is one of the easiest judgments in the whole review. The thesis can legitimately be broader: repository evolution, exploratory notebooks, tool comparisons, field workflows, DINOv2 and GEE experiments, and future CHM integration all fit naturally there. The paper should be narrower and more disciplined: repeated UAV RGB orthomosaics, alignment, graph-based crown identity, consensus crowns, phenology signal extraction, and carefully chosen validation. [11](#)

## J. Validation and experiment plan

“What experiments would make this convincing?” [filecite\turn0file1](#)

`main.md` documents detection metrics, alignment defaults, graph reports, partial chains, consensus crowns, species splits, and satellite classifier experiments. [filecite\turn0file0](#)

This is supportable, but the files also show where the plan must improve. A convincing paper needs at least detection evaluation on a labelled subset, alignment ablation, chain-quality evaluation on hand-linked crown subsets, consensus-versus-single-date geometry comparison, phenology-signal validation against field labels or expert review, and a clean downstream demonstration. RandCrowns is particularly relevant if polygon labelling uncertainty is non-trivial. MatchPlant is useful as a nearby example of reusing detected geospatial instances across time for trait extraction. [12](#)

## K. Specific positioning statement

“Give me a crisp contribution statement” [filecite\turn0file1](#)

“The final scientific object is not just a crown detection...” [filecite\turn0file0](#)

The crispest statement is: **We present an end-to-end UAV RGB pipeline that turns repeated orthomosaics into temporally stable individual-tree crown identities and crown-level phenology trajectories through multi-threshold crown delineation, residual post-alignment, graph-based association, and consensus crown generation.** That is the shortest accurate sentence that captures both files and the project’s strongest novelty. [filecite\turn0file0](#)

The ten specific doubts in `prompt.md` resolve in the same direction. The project is enough for a thesis; the central contribution is integrated tracking-plus-phenology rather than detection alone; tracking is central; satellite work is secondary; the rule-based score should be framed as a baseline or heuristic unless strongly validated; the overall pipeline is coherent if narrowed; the weakest links are validation and overbreadth; the most valuable added experiments are alignment and tracking ablations plus phenology QA; the way to avoid “just using existing tools” is to foreground the unsolved temporal-identity problem; and the best single contribution line is the one given above. [filecite\turn0file1](#)  
[filecite\turn0file0](#)

## Assumptions, ambiguities, conflicts, and suggested prompt revisions

The files are rich, but not perfectly closed under all the claims a paper would want to make. The necessary assumptions should therefore be stated openly.

The most important assumptions are these. The active core method uses RGB orthomosaics rather than multispectral or raw-frame reconstruction as its main phenology substrate. Field labels are important but appear partial, and in the material reviewed they look stronger for species and general validation than for exhaustive crown-by-date phenophase truth. Satellite, DINOv2, and species-classification work are real components of the repository, but they are downstream rather than core to the longitudinal identity contribution. Finally, no full tracking gold standard was explicitly surfaced in the file summaries, so any strong re-identification claim will need either a labelled subset or carefully defined proxy metrics.

A few ambiguities or conflicts deserve explicit revision before submission.

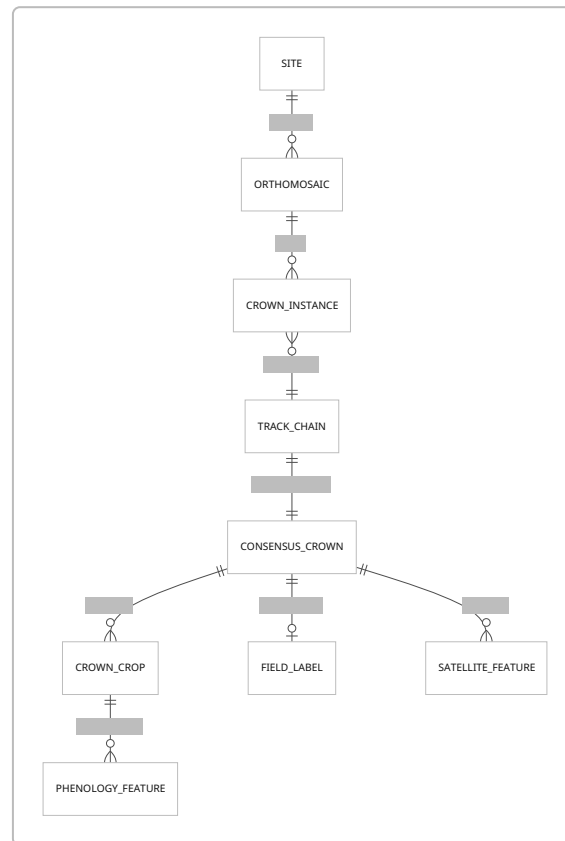
| Issue                                          | Why it matters                                                                | Evidence                                                                                                                                                            | Recommended clarification                                                                           |
|------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Phenology threshold inconsistency              | A paper cannot quietly use different defaults for the same leaf-state concept | <code>main.md</code> notes a 0.85 default in the analyser class and a 0.70 CLI default in the active phenology lane<br><code>filecite:turn0file0</code>             | State one default, justify it, and report a sensitivity analysis                                    |
| CHM / DTM / DSM mention versus active pipeline | Readers may think canopy height is already an evaluated component             | <code>main.md</code> says companion-paper context mentions it, but active pipeline does not yet compute it<br><code>filecite:turn0file0</code>                      | Move CHM to future work unless results exist                                                        |
| Detection quality versus paper ambition        | Low recall will weaken any claim that starts and ends at crown delineation    | Early report metrics in <code>main.md</code> show low recall and many false positives despite decent IoU when correct<br><code>filecite:turn0file0</code>           | Make detection an enabling module, not the headline contribution                                    |
| Orthomosaic artefacts                          | Reviewers will ask whether residual drift and stitching affect biology        | <code>main.md</code> warns of a bad LHC Dec-9 orthomosaic; literature also flags orthomosaic limits<br><code>filecite:turn0file0</code> <sup>13</sup>               | State exclusions and orthomosaic limitations clearly                                                |
| Over-broad scope                               | Too many side stories dilute the contribution                                 | Both files include species, satellite, flowering, and future ideas beyond the core pipeline<br><code>filecite:turn0file1</code><br><code>filecite:turn0file0</code> | Keep the paper centred on identity-preserving phenology; move extras to thesis chapters or appendix |

The prompt itself would improve with a few small revisions. It should explicitly define whether the requested output is mainly for a **thesis chapter strategy** or for a **journal-paper framing**, because the best answer differs slightly. It should also distinguish between “field validation”, “species labels”, and



“phenophase labels”, because those are not interchangeable. Finally, it should say whether the goal is to defend the current rule-based phenology lane as a final method, or to position it as a baseline awaiting more labels. Those three edits would remove most of the residual ambiguity. [filecite turn0file1](#)  
[filecite turn0file0](#)

An entity-relationship view helps show the core logic that should stay central:



Open questions remain, but they are mostly validation questions rather than conceptual ones. The files do not clearly expose the size of the final labelled tracking subset, the exact scope of phenophase labels, or the final paper-ready quantitative summary for alignment plus tracking. Those should be treated as limitations unless additional internal material exists outside the two files.

[filecite turn0file0](#)

## Draft response ready to submit

Below is a complete draft answer that could be submitted as the response to `prompt.md`, grounded in the two files and supplemented with primary or official external references where they materially strengthen the framing.

### Working recommendation

The project is strong enough for a thesis, and it can plausibly support a paper if it is framed correctly. The right central framing is **not** “tree crown detection from UAV orthomosaics”. It is **individual-tree phenology monitoring from repeated UAV RGB surveys, where the key technical contribution is turning unstable per-date crown detections into temporally stable tree identities and crown-level phenology trajectories**. That framing is directly supported by the repository map, which identifies

temporal identity as the hard problem and stable tree identity as the final scientific object.

Suggested title options

| Option                                                                                                                                | Best use                                    | Why it works                              |
|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|-------------------------------------------|
| <i>Tracking Individual Tree Phenology from Repeated UAV RGB Orthomosaics Using Graph-Based Crown Association and Consensus Crowns</i> | Best paper title                            | Precise and method-centred                |
| <i>From Crown Detection to Tree Identity: A UAV RGB Pipeline for Individual Tree Phenology Monitoring</i>                             | Best thesis chapter title                   | Communicates the conceptual shift clearly |
| <i>Drone-Based Individual Tree Phenology Monitoring Through Temporal Crown Tracking and Consensus Geometry</i>                        | Best thesis title                           | Slightly broader and more thesis-like     |
| <i>Repeated UAV RGB Surveys for Tree-Level Phenology: Alignment, Crown Tracking, and Downstream Ecological Inference</i>              | Best if you want a bridge to satellite work | Keeps satellite utility as downstream     |

My strongest recommendation for the paper is the first title, and for the thesis as a whole the third.

Abstract draft for a paper

Repeated UAV RGB surveys make it feasible to observe canopy change at the scale of individual trees, but converting per-date orthomosaics into biologically meaningful phenology trajectories remains difficult because crown delineations vary across time due to georeferencing drift, photogrammetric artefacts, illumination differences, and segmentation instability. We present an end-to-end pipeline for individual-tree phenology monitoring from repeated UAV RGB orthomosaics that combines multi-threshold crown delineation, residual orthomosaic post-alignment, graph-based temporal crown association, gap-filling, and consensus crown generation to produce temporally stable tree identities. For each consensus crown, we extract per-date crown crops and RGB-derived phenology features, and we link these outputs to field and QField annotations where available. The framework is designed to move beyond single-date crown detection by prioritising temporal identity preservation and crown-level time-series consistency. We position downstream satellite and embedding experiments as secondary demonstrations of utility rather than the main methodological contribution. This framing is appropriate because existing literature is rich in crown delineation methods, whereas robust identity-preserving tree-level phenology from repeated UAV orthomosaics remains less standard and more operationally challenging.

Abstract draft for a thesis

This thesis investigates how repeated UAV RGB surveys can be transformed into tree-level phenology records for individual crowns observed across time. Although orthomosaics and crown delineation models make single-date tree mapping increasingly accessible, long-term ecological monitoring requires more than detection: it requires stable identity assignment across dates, despite drift, shape change, detection failure, and photogrammetric inconsistency. I develop a workflow that processes repeated UAV imagery into georeferenced orthomosaics, derives crown polygons with Detectree2, improves temporal comparability through post-alignment, performs graph-based crown association

across dates, and constructs consensus crown geometries that support per-tree RGB phenology time series. The resulting crown identities can be linked to field and QField observations and used in downstream analyses, including exploratory satellite and embedding-based classification tasks. The main contribution of the thesis is therefore the design and evaluation of a practical pipeline for converting repeated UAV orthomosaics into temporally stable individual-tree phenology objects, rather than merely improving single-date crown detection. [filecite turn0file0](#) 15

### What the central thesis should be

The central thesis should be:

Repeated UAV RGB orthomosaics can support individual-tree phenology monitoring if crown detections are converted into temporally stable tree identities through explicit alignment, graph-based temporal association, and consensus crown construction.

That is the best synthesis of the two files. It is technically faithful, scientifically meaningful, and much stronger than a “we used Detectree2 on UAV orthomosaics” narrative. [filecite turn0file0](#) [filecite turn0file1](#)

### Related work map

The related work should be organised into five clusters.

First, **individual tree crown detection and delineation from aerial imagery**. A recent review of ITCD from optical remote sensing shows that the field is already broad and increasingly dominated by deep learning. Detectree2 is directly relevant because it is an official tree-crown instance-segmentation framework built on Detectron2 and Mask R-CNN. DeepForest is a relevant open-source ecological baseline, though its role is better treated as historical or comparative in your story. Recent UAV RGB studies also show that high-quality annotations can yield very strong crown segmentation performance. 16

Second, **limitations of crown delineation and evaluation**. This literature matters because it tells you not to overclaim. RandCrowns exists specifically because crown labels are imprecise and polygon evaluation is noisy. More importantly, recent TLS-based validation work found that widely used aerial crown models such as DeepForest and Detectree2 can look dramatically worse when evaluated against independent high-fidelity labels rather than hand-labelled aerial polygons. That result is extremely useful for your framing: it supports moving the paper away from a crown-detection performance story and towards a longitudinal identity-and-phenology story. 17

Third, **registration and temporal comparability**. The scikit-image implementation of `phase_cross_correlation` is built on established subpixel registration work and explicitly provides efficient subpixel translation registration, with the usual trade-offs between phase normalisation, illumination robustness, and noise sensitivity. This is a solid basis for justifying your translation-focused post-alignment stage. At the same time, remote sensing registration reviews show that registration remains hard and that no single method dominates all cases, which is a good reason to present PCC-style alignment as a pragmatic design choice rather than a universal solution. Official OpenDroneMap documentation also distinguishes relative from absolute accuracy and documents multi-temporal accuracy workflows, which strengthens the argument that residual alignment correction is a real problem after reconstruction. 18

Fourth, **temporal association and phenology extraction**. Here the literature is thinner in your exact niche, which is good for novelty but demands careful honesty. MatchPlant is a useful nearby example from UAV plant phenotyping because it reuses detected plant instances across time points to extract traits. There is also fresh work showing that image-based phenology at the individual level remains challenging even with modern vision models. Together these help justify why your graph-based temporal crown association matters scientifically. <sup>19</sup>

Fifth, **downstream satellite and embedding utility**. Sentinel-2 officially provides 13 bands at 10 m, 20 m, and 60 m spatial resolutions with free access and a 5-day revisit design, and Level-2A surface reflectance is the analysis-ready product. Google Earth Engine is an official geospatial processing service for large-scale analysis. DINOv2 provides strong self-supervised visual features across tasks and image domains. These sources justify a secondary downstream lane, but they also make clear that this lane is conceptually distinct from the core UAV temporal-identification contribution. <sup>20</sup>

### What is actually novel here

The novelty is the **integration around temporal identity**, not any one component in isolation. Detectree2 is existing. Mask R-CNN is existing. WebODM and OpenDroneMap are existing. RGB greenness features are existing. Earth Engine and DINOv2 are existing. If the paper reads as a sequence of tool applications, reviewers will quite reasonably push back. <sup>21</sup>

What is novel and defensible is that the system addresses a real ecological bottleneck: how to turn noisy, changing, sometimes missing crown delineations from repeated orthomosaics into stable per-tree objects that can support phenology analysis. The multi-threshold crown storage, residual temporal alignment, graph-based matching, gap-filling, and medoid-like consensus geometry all contribute directly to that bottleneck. That is a better novelty statement than any “best segmentation” claim.

filecite turn0file0

### How to defend the key design choices

Orthomosaics should be defended as the practical substrate for repeat-site ecological GIS work, not as an assertion that raw imagery is always superior. In fact, recent work on palm detection shows that raw imagery can outperform orthomosaics in some deployment scenarios, while orthomosaics still retain practical value for generalisation and downstream mapping. That gives you the right tone: orthomosaics are useful and operationally sensible, but they introduce artefacts that justify your alignment and consensus stages. <sup>13</sup>

Detectree2 is defensible because it is explicitly a tree-crown delineation framework built for aerial imagery, based on a strong instance-segmentation backbone. The right phrasing is not “Detectree2 solves crown delineation”, but “Detectree2 provides the per-date crown candidates that our temporal identity pipeline refines into stable tree objects.” <sup>22</sup>

Graph matching is defensible because the repository history itself shows why Hungarian one-to-one assignment is too brittle: splits, merges, and missing detections break local pairwise matching. Your graph-based stage is therefore not gratuitous complexity; it is the mechanism that lets the pipeline represent longitudinal uncertainty and continuity. filecite turn0file0

Medoid or consensus crowns are defensible because they create a stable spatial object for downstream crop extraction and time-series analysis. Without consensus geometry, every per-date phenology

measurement is contaminated by per-date shape noise. That is one of the strongest parts of the current conceptual framing. [filecite turn0file0](#)

Rule-based phenology scores should be defended as **transparent, reproducible, low-data baselines**. They are useful precisely because they do not require heavy annotation. But unless you have extensive phenophase labels, they should not be sold as final biological truth. Recent image-based phenology work supports the idea that useful phenological status can be estimated from imagery without full supervised learning, but it also underlines that individual-level monitoring is hard. That combination argues for positioning your score as a first-pass phenology signal or weakly supervised baseline. [23](#)

Downstream satellite classifiers should be defended as **evidence that the crown-level time series and validation pipeline produce useful training or interpretation objects**, not as the paper's main contribution. Official Sentinel-2, Earth Engine, and DINOv2 sources are enough to justify why those experiments are interesting. They are just not the cleanest centrepiece for this story. [24](#)

### What the paper should claim and what it should not claim

It should claim that the main contribution is a pipeline for constructing temporally stable crown identities and crown-level RGB phenology trajectories from repeated UAV orthomosaics. It should claim that alignment and temporal association are necessary because per-date crown delineations are unstable. It should claim that consensus crowns improve the downstream unit of analysis. It should claim that the resulting objects can be linked to field/QField validation and can support downstream experiments. All of those claims are well grounded in the files. [filecite turn0file0](#)

It should not claim that the method produces comprehensive species identification unless the species-labelled evaluation is strong. It should not claim that canopy-height modelling is already a core result if CHM is not yet active in the main pipeline. It should not claim that the rule-based leaf-state score is a final ecological truth unless you can validate it at scale. It should not bury orthomosaic artefacts or the need to exclude known bad mosaics. [filecite turn0file0](#)

### The best paper structure

A strong paper structure would be:

**Introduction.** Motivate individual-tree phenology, repeated UAV surveys, and the gap between single-date crown delineation and longitudinal tree identity.

**Related work.** Cover crown delineation, registration, temporal association, image-based phenology, and downstream satellite/embedding analysis. Use this section to argue that detection is mature, but longitudinal identity preservation is under-served. [25](#)

**Study system and data.** Explain repeat UAV surveys, orthomosaics, and field/QField context. Keep site details concrete but concise. [filecite turn0file0](#)

**Method.** Present the pipeline in order: orthomosaics; multi-threshold Detectree2 crowns; post-alignment with `pcc_tiled`; graph-based crown association; chain assembly and gap-filling; consensus crowns; crop generation and RGB phenology features. [filecite turn0file0](#)

**Validation and ablations.** This section is critical. Include detection subset evaluation; alignment ablation; tracking subset evaluation; consensus-versus-no-consensus comparison; and phenology signal QA. If annotation uncertainty is large, cite RandCrowns. <sup>26</sup>

**Discussion.** Emphasise why orthomosaic-based phenology needs identity-preserving tracking, what the approach does well, and where it still depends on manual QA or limited labels.

**Downstream utility.** Put satellite, DINOv2, and species-classifier experiments here or in the appendix. They are interesting, but they should not fracture the main narrative. <sup>27</sup>

**Conclusion.** Restate the key contribution as a practical, identity-preserving UAV phenology pipeline.

### What would make the work much stronger

If you can add just a few things, add these. Create a manually linked temporal subset for a few sites and dates so that tracking accuracy can be reported directly. Run an alignment ablation comparing no post-alignment, PCC, tiled PCC, and any plausible affine baseline on the same subset. Compare consensus crowns against direct per-date crop extraction for phenology feature stability. Where annotation noise is obvious, report an uncertainty-aware evaluation view, not just IoU. And if field phenology labels are limited, explicitly present the rule-based phenology lane as a baseline awaiting richer labels. Those additions would materially improve reviewer confidence. <sup>28</sup>

### The weakest links and how to discuss them

The weakest links are not fatal, but they must be handled carefully. First, crown delineation quality is visibly imperfect in the early report, so the paper should not be positioned as a segmentation benchmark. Second, orthomosaic artefacts are real and should be stated plainly. Third, tracking needs stronger quantitative validation than a software pipeline usually provides by default. Fourth, rule-based phenology thresholds should be harmonised and sensitivity-tested. Fifth, satellite and species lanes should not overtake the central claim. If you acknowledge these issues yourself, they become limitations rather than surprises. <sup>29</sup>

### Direct answers to the major doubts

Yes, this is enough for a thesis, because the combined problem of repeated-site UAV phenology, temporal crown identity, and downstream validation is substantive enough.

The main contribution is the integrated **tracking plus consensus plus phenology** pipeline. Detection alone is not enough.

Tracking is central. In fact, it is the intellectual centre of the project.

The satellite side should be secondary in the main paper. It fits naturally as downstream utility, extension material, or a separate paper.

The rule-based score should be treated as a transparent heuristic baseline unless you can validate it strongly.

The pipeline does read as a coherent contribution, but only if you centre it on stable tree identity over time.

The weakest links are validation, threshold consistency, orthomosaic artefacts, and scope creep.

The experiments that would most improve the work are direct tracking-ground-truth subsets, alignment ablations, consensus ablations, and phenology QA.

To avoid the “just using existing tools” criticism, never describe the work as WebODM plus Detectree2 plus some features. Describe it as a system for solving temporal crown identity and building stable phenology objects from repeated UAV orthomosaics.

The crispest contribution statement is this:

We present an end-to-end UAV RGB pipeline that converts repeated orthomosaics into temporally stable individual-tree crown identities and crown-level phenology trajectories using multi-threshold crown delineation, residual post-alignment, graph-based temporal association, and consensus crown generation.

### Final recommended framing

If I had to choose one sentence to guide the whole write-up, it would be this: **the thesis is about converting repeated UAV orthomosaics into stable tree identities for phenology analysis, not about detecting crowns in isolated aerial images.** That is the framing most strongly supported by both files, the most defensible against the literature, and the one that gives the work a real methodological contribution rather than the appearance of a toolchain exercise. filecite $\rightarrow$ turn0file1 $\rightarrow$ filecite $\rightarrow$ turn0file0 $\rightarrow$  30

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1 2 5 6 14 16 25 30 <https://arxiv.org/abs/2310.13481>  
<https://arxiv.org/abs/2310.13481>

3 7 15 21 22 <https://patball1.github.io/detectree2/>  
<https://patball1.github.io/detectree2/>

4 9 <https://docs.opendronemap.org/>  
<https://docs.opendronemap.org/>

8 <https://arxiv.org/abs/2503.14273>  
<https://arxiv.org/abs/2503.14273>

10 13 29 <https://arxiv.org/abs/2509.12400>  
<https://arxiv.org/abs/2509.12400>

11 <https://arxiv.org/abs/2304.07193>  
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18 28 <https://scikit-image.org/docs/stable/api/skimage.registration.html>  
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20 24 27 <https://dataspace.copernicus.eu/explore-data/data-collections/sentinel-data/sentinel-2>  
<https://dataspace.copernicus.eu/explore-data/data-collections/sentinel-data/sentinel-2>

<sup>23</sup> <https://arxiv.org/abs/2603.07817>

<https://arxiv.org/abs/2603.07817>